

Abstract No.

Ferritin nanoparticles as strategic delivery tools for human organotypic brain slices: a proof-of-concept study

Zaira Spinello¹, Giulia Villa², Daniela Nardozi³, Natalia Pediconi⁴, Martina Brunetti³, Chiara Cappelletti⁵, Anna Citarella¹, Kevin Joseph^{6,7}, Alessio Incocciati⁵, Francesca Gianno⁵, Claudia Palazzo⁸, Maria Teresa Viscomi^{8,9}, Vatsal D. Jariwala^{6,7}, Nicolas Neidert⁷, Roland Rölz⁷, Jurgen Beck⁷, Zein Mersini Besharat³, Agnese Po³, Manila Antonelli³, Alessandra Bonamore³, Alberto Macone³, Oliver Schnell¹⁰, Laura Masuelli³, Alberto Boffi³, Vidhya Madapusi Ravi^{6,7}, Dieter Henrick Heiland^{11,12}, Elisabetta Ferretti⁴, and Giuseppina Catanzaro^{1,}*

¹Link Campus University

²Friedrich-Alexander University Erlangen Nuremberg, Erlangen, Germany.

³Sapienza University of Rome, Rome, Italy

⁴Sapienza University of Rome

⁵Sapienza University, Rome, Italy

⁶Medical Center-University of Freiburg, Germany

⁷Medical Center-University of Freiburg, Freiburg, Germany

⁸Università Cattolica del Sacro Cuore, Rome, Italy

⁹Fondazione Policlinico Universitario A. Gemelli, Rome, Italy

¹⁰Medical Center - University of Freiburg, Freiburg, Germany

¹¹Friedrich-Alexander University Erlangen Nuremberg, Erlangen, Germany

¹²University Clinic Erlangen, Friedrich-Alexander-Universität-Erlangen, Erlangen-Nürnberg, Germany

*Presenting author

Introduction:

Glioblastoma (GB) remains one of the most aggressive primary brain tumors despite multimodal treatment approaches. Although microRNA (miRNA)-based replacement therapies hold considerable therapeutic potential, their clinical application is hindered by poor stability, limited tissue penetration, and insufficient tumor-selective delivery. The transferrin receptor CD71, encoded by TFRC, is a key mediator of iron uptake and is highly expressed in metabolically active GB cells, thereby representing a potential endogenous gateway for targeted delivery.

Methods:

To characterize TFRC distribution within GB tissue, we integrated single-cell reference atlases with spatial transcriptomic analyses. CD71 protein expression was evaluated by immunohistochemistry, while miR-139-5p localization was assessed using miRNA scope in matched tumor and adjacent peritumoral regions. miRNA delivery was performed using an engineered humanized ferritin nanocage derived from *Archaeoglobus fulgidus* and loaded with a PAMAM dendrimer (hFt-NP). Nanoparticle uptake, tissue penetration, miR-139-5p restoration, and biological effects were investigated in patient-derived GB cell cultures and organotypic human cultures (OHCs).

Results:

TFRC expression was associated with malignant astrocytic, hypoxic, and inflammatory transcriptional programs, with TFRC-high tumor cells preferentially localized within myeloid-rich niches. At the protein level, CD71 was markedly enriched in tumor regions, whereas miR-139-5p expression was predominantly retained in peritumoral areas. hFt-NPs efficiently penetrated OHCs, displayed CD71-associated uptake, restored miR-139-5p levels, reduced Ki-67 positivity, and promoted progressive



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cell-cycle arrest.

Conclusions:

Our findings identify TFRC/CD71-high niches as biologically relevant targets for ferritin-mediated miRNA delivery in GB. In addition, these observations establish patient-derived OHCs as a clinically relevant ex vivo platform for studying nanoparticle penetration, targeted delivery, and therapeutic activity within preserved tumor microenvironmental contexts.



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